Yusra Fatima

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EDUCATION

Christ University, Delhi NCR Campus Expected Graduation: 2029

B.Sc. Data Science and Artificial Intelligence (2BDA) — CGPA: 3.7/4.0

Relevant Coursework: R Programming, Python, Data Structures & Algorithms, Machine Learning, AI Foundations

Indian Institute of Technology (IIT) Concurrent

B.Sc. in Machine Learning (parallel program)

TECHNICAL SKILLS

Languages: Python, R, SQL

Tools & Platforms: Git, Google Colab, Jupyter Notebooks

Core CS: Data Structures & Algorithms, Graph Theory, Object-Oriented Design

AI/ML: Machine Learning, Deep Learning, Hidden Markov Models, Generative AI, Agentic AI Workflows

Quant/Math: Probability & Bayesian Methods, Riemannian Geometry, Time-Series Analysis

Interview Prep: 6 months of dedicated practice in LeetCode (DSA), System Design, ML, and DL

EXPERIENCE

Freelance AI Developer 2026 – Present

Independent Contractor (Agentic AI)

Built a customer support chatbot for a freelance client using agentic AI workflows, automating responses to common customer queries.

STARTUPS

Founder 2026 – Present

Surgeon-Facing AI Aesthetic Consultation Platform

Designing a B2B platform combining 3D facial scanning, facial attribute analysis, and surgical recommendation generation for plastic surgeons, supported by a referral network.

Developing a proprietary mathematical framework using Riemannian geometry and Bayesian inference to quantify facial attractiveness across diverse ethnicities, with an ethnicity-first, evergreen approach to beauty standards.

Identified a closed surgeon-data feedback loop as the platform's core competitive moat, differentiating from existing players such as Qoves Studio.

Compiled a 40-paper literature database spanning facial geometry, cultural aesthetics, AI beauty scoring, and probabilistic face models to ground the platform's research foundation.

RESEARCH

Mathematical Framework for AI-Driven Aesthetic Analysis In Progress

Authoring a research paper developing a mathematical framework grounded in Riemannian geometry and Bayesian attractiveness modeling, underpinning the aesthetic consultation platform above.

PROJECTS

A-CMSI: Adaptive Composite Market Stress Index

Built a Hidden Markov Model-based index to detect regime shifts and market stress using historical price data for AAPL, MSFT, and SPY.

Conducted literature review and benchmarking against existing volatility and stress indices.

Pathfinding Visualizer

Built an interactive grid-based visualizer animating Dijkstra's, A*, BFS, and DFS algorithms to compare pathfinding strategies in real time.

Trie-Based Autocomplete & Search Engine

Implemented a Trie data structure for prefix-based autocomplete, extended with frequency-based ranking and edit-distance typo tolerance.

Mini Key-Value Store

Built a lightweight key-value database implementing B-tree/hashmap indexing and an LRU cache for fast lookups.

Huffman Coding File Compression Tool

Implemented Huffman coding from scratch using greedy algorithms, priority queues, and binary trees; benchmarked compression ratios against real files.

Task Scheduler & Dependency Resolver

Built a topological sort-based dependency resolver with cycle detection, modeled as a minimal build system.

Rate Limiter & Load Balancer Simulator

Simulated sliding window and token bucket rate-limiting algorithms to model time-based request throttling.

AI-Generated Fake Review Detector/Generator 1st Year

Built a generative AI tool to produce synthetic product reviews, exploring text generation techniques and review authenticity patterns.

LEADERSHIP & ACTIVITIES

Spoken Word Poetry — 2nd Place, Intra Blossoms '26

Awarded 2nd place in a university-wide spoken word poetry competition.